RDFlow: monitoring data interoperability and ontology evolution for rare diseases

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Abstract

Practitioners working with biomedical semantic resources often need to understand how existing data are structured, how modeling patterns appear in practice, and how structure evolves over time. In this demo, we present an application of RDFlow to HOOM, an ontological module developed by Orphanet to relate concepts from disease (ORDO) and phenotypes (HPO). Starting from OWL/XML representations made available by Orphanet, we apply a reproducible semantic workflow that produces a structure-oriented RDF projection of the resource. This representation supports data transformation, schema extraction, and the exploration, visualization, and comparison of ontology structure, for example through tools such as SheXer, rdf-config, and Rudof. The demo illustrates how this workflow facilitates structural understanding and evolution analysis of real-world semantic resources in a concrete and reusable environment.

Keywords

RDF transformation pipeline, SheXer, Shape-based comparison, Knowledge graph understanding, rdf-config, Rudof

1. Introduction

In many life science projects, ontologies are reused as existing resources rather than engineered by the teams that integrate them. Practitioners interacting with such resources frequently need to answer structural questions: Which kinds of entities are present? How are they typically connected? What changes between versions or across sources? These questions are central to integration, maintenance, and reuse tasks, yet they are not the primary focus of standard ontology representations. [1, 2].

Semantic resource serializations such as OWL are designed to preserve formal semantics and support interoperability, but they are not always suitable for direct structural inspection or comparison [3]. Axioms are often encoded in complex RDF patterns that are difficult to summarize or analyze systematically. As a result, understanding the actual structure of a resource often requires substantial manual effort or custom querying.

This demo addresses this practical gap by applying RDFlow [4] to derive explicit, reusable structural representations of an existing ontology and by using shapes as descriptive summaries of observed modeling patterns.

2. Use Case: HOOM

HOOM (HPO-ORDO) [5] is a semantic module developed by Orphanet to represent relationships between rare diseases described in ORDO (Orphanet Rare Disease Ontology)[6] and phenotypic features

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described in the Human Phenotype Ontology (HPO) [7]. It supports semantic interoperability across rare diseases and phenotype resources.

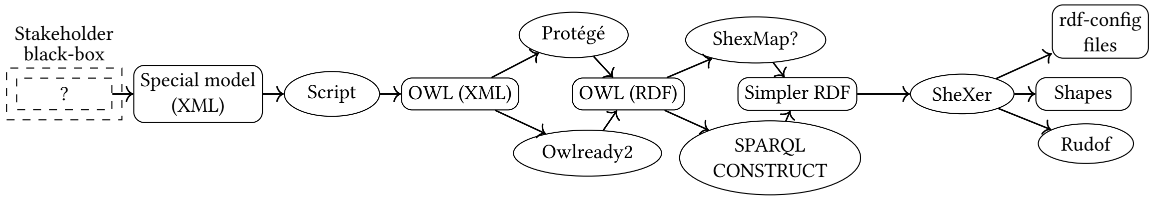
HOOM is primarily TBox-oriented and is distributed in several formats, including OWL/XML and XML-based representations made accessible through Orphanet services. This demo focuses on the OWL/XML representation that is readily available and commonly reused.

This use case is particularly relevant because the structure of HOOM is not directly analyzable due to complex OWL axiom encodings and the absence of a rich ABox, thus requiring explicit transformations to reveal structural patterns.

3. RDFlow Pipeline and Structural Outputs

The demo materializes RDFlow principles, with each step explicitly represented and aligned with the pipeline shown in Figure 1. The goal is not to alter the semantics of HOOM, but to derive a structural RDF graph suitable for analysis.

Figure 1: General RDFlow pipeline illustrated through the HOOM application.



The pipeline consists of the following steps:

- **Source representation** : The ontology is obtained in OWL/XML format [8].
- **OWL/RDF normalization** : The OWL/XML source is converted into an OWL/RDF representation using tools such as Protégé [9] or Owlready2 [10]. This step ensures compliance with the OWL 2 mapping to RDF and makes the underlying RDF graph explicit. [3]
- **Structural RDF projection** : A simplified, analysis-oriented RDF graph is derived using SPARQL CONSTRUCT queries. This projection captures entity categories, main relations, and recurring structural patterns, while abstracting away from complex OWL axiom encodings. The resulting graph is an explicit artefact that can be reused, versioned, and compared.
- **Shape extraction** : RDF shape languages are used to express structural schemas that summarize how entities are typically structured and connected. These schemas are automatically derived from the structural RDF graph, for instance using SheXer [11], which produces ShEx[12]/SHACL[13] shapes as well as RDF-Config schema representations to support subsequent analysis steps.
- **Exploration and comparison** : The RDF-Config files obtained enable visualizations, supporting understanding and comparison of ontology structure [14]. Some other tools such as RuDoF can be used to formalize selected patterns or support quality-oriented analyses [15].

This pipeline enables the full exploitation of shape-based tools by providing them with inputs that reflect the structural characteristics of the ontology.

4. Conclusion and Perspectives

This demo shows how RDFlow can be applied to an existing rare disease ontology to produce explicit, reusable structural representations and shape-based summaries. By making ontology structure observable and comparable, the approach supports practical tasks such as exploration, comparison, and evolution analysis. This contributes to improving the findability, interoperability, and reusability of semantic resources in line with FAIR principles [16] .

Future work includes evaluating direct transformations from XML to RDF projections using existing technologies such as ShExML [17], applying the pipeline to multiple HOOM releases or other biomedical ontologies, and further exploring the use of shapes for ontology curation and interoperability monitoring.

Declaration on Generative AI

During the preparation of this work, the authors used Chat-GPT-4 and DeepL in order to: Grammar and spelling check. After using these tools/services, the authors reviewed and edited the content as needed and takes full responsibility for the publication's content.

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